

COURSE TITLE : CNC PROGRAMMING
COURSE CODE : 5302
COURSE CATEGORY : A
PERIODS / WEEK : 4
PERIODS / SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction to computer numerical control	13
2	Program Format, Programming Procedure For Manual CNC Programming Preparatory Functions And Miscellaneous Functions	16
3	CNC Lathe Part Programming	16
4	CNC Milling Part Programming	15
TOTAL		60

COURSE OUTCOME:

At the end of the course the trainee will become able to:-

1. Understand numerical control systems and the advantages of CNC Machines.
2. Identify the need for CNC.
3. Distinguish various axis motion and direction.
4. Determine machine zero, work zero and tool reference points.
5. Understand the part programming and familiar with M codes and G codes.
6. Become familiar with various cutting tools and their specifications
7. Prepare part programme for turning and milling jobs
8. Calculate the speed, feed and depth of cut for CNC machining.
9. Understand the effect of tool and cutter radius and give necessary compensation.
10. Prepare part programs using subprograms and machining cycles
11. Do thread cutting on lathe and milling machine.
12. Write programs using mirror images.
13. Edit the programme and check the tool path by Simulation.

MODULE I

1.1.0 Introduction to Computer Numerical Control.

- 1.1.1 Distinguish between conventional and CNC machines.
- 1.1.2 Understand Advantages and Disadvantages of CNC.
- 1.1.3 Identify the Applications and Need for CNC.

1.2.0 Understand the main parts of CNC machine.

- 1.2.1 Identify the different Components of CNC system-part programme, computer-MCU and processing machine.
- 1.2.2 Understand AC or DC feed motors, re-circulating ball screws, spindle speed, stepper motors and servo motors.

1.3.0 Understand Data Processing In CNC System.

- 1.3.1 Explain Data reading, conversions of data-binary to decimal & BCD.

1.4.0 Identify different directions of Axis movement and Co-ordinate systems.

- 1.4.1 Understand Linear axes designations X Y Z.
- 1.4.2 Explain Rotary axis ABC and right and thumb rule.
- 1.4.3 Distinguish between Cartesian system and Polar co-ordinate system.
- 1.4.4 Describe Incremental and Absolute systems of programming.

MODULE II

2.1.0 Understand about the Reference points.

- 2.1.1 Illustrate Program zero, machine zero, and tool reference points

2.2.0 Understand the Preparatory Functions and Miscellaneous Functions.

- 2.1.1 EXPLAIN G CODES and M CODES.

2.3.0 Explain the Program Format.

- 2.3.1 List the meaning of address characters, word-address structure, and program block and block numbers.
- 2.3.2 Explain Programming Procedure For Manual CNC Programming.
- 2.3.3 Define the machining operations, necessary tools, Technological & Geometric data and Generating CNC program test.

MODULE III

3.1.0 Understand the effect of Tool Nose Radius Compensation.

- 3.1.1 Explain various turning tools, Compensation using G41, G42, G40-Cancel Compensation.

3.2.0 Describe the fundamentals of Machining cycles (FIXED/CANNED CYCLES)

- 3.2.1 Understand the definition, Purpose, Format and examples of Machining cycle.
- 3.2.2 Prepare part Programs using Machining cycles.

3.3.0 Understand part programme entry and execution.

- 3.3.1 Become familiar with Programme editing, Checking syntax, Execution and Simulation of tool path.

MODULE IV

4.1.0 Distinguish various axis motions, directions and reference points.

- 4.1.1 Understand about machine axes, 4axis, 5axis and 6axis.
- 4.1.2 Explain Machine zero, Work zero and Reference zero or or Home position.

4.2.0 Understand the Preparatory Functions and Miscellaneous Functions.

- 4.2.1 Explain G CODES and M CODES.

4.3.0 Write part programs for Point-Point Machining and Contour Milling

- 4.3.1 Prepare part programs using Linear and Circular interpolations- G01/G02/G03 and XYIJKR.

4.4.0 Understand Programming examples using Machining cycles (CNC FIXED/CANNED CYCLES).

- 4.4.1 Explain Drilling, Tapping and Boring cycles.
- 4.4.2 Describe the Drilling Cycles G73, G81, G82, G83 and Programming examples.
- 4.4.3 Explain canned cycle cancel G80, Z return code G99, G98 and Dwell.
- 4.4.4 Explain Boring and Tapping Cycles, Format and do programming examples.

- 4.4.5 Describe Milling Rectangular Pockets and Milling Circular Pockets.
- 4.4.6 Explain format, position of pocket, selection of starting corner, Feed rate, spindle speed, Milling direction, pocket, radius, feed rate, spindle speed, Finishing allowance and Programming examples.
- 4.5.0 Understand the effect of Cutter Radius Compensation, Zero Offset and Length Compensation.**
- 4.5.1 Effect of cutter radius on contour, Tool length compensation and CRC left or right side.
- 4.6.0 Describe programming using Sub programs and Mirror Images.**
- 4.6.1 Explain or define, Prerequisites and do Programming using M98, M99, M70, M71, M80 & M81.
- 4.7.0 Understand part programme entry and execution.**
- 4.7.1 Become familiar with Programme editing, Checking syntax, Execution and Simulation of tool path.

CONTENT DETAILS

MODULE I

Introduction to Computer Numerical Control- Difference between conventional & CNC machines-advantages of CNC- Applications-Need for CNC-Components of CNC system-part programme-computer-Machine Control unit-processing machine -Data Processing In CNC System- Data reading, conversion of data-binary, decimal, & BCD, Axes Designation X Y Z AXES, directions of axis movement- Rotary axis ABC-right and thumb rule-The Co-Ordinate System---Position, Planes, Cartesian System- Polar Co-Ordinates-Main Parts Of CNC machine – axis system-Point to point contouring system, incremental and absolute system-DC feed motors, circulating ball screws, spindle speed, stepper motors.

MODULE II

REFERENCE POINT- Program zero, machine zero, and tool reference point-Preparatory Functions And Miscellaneous Functions-Explanation Of G Codes-Explanation Of M Codes- Program Format-Address characters, word, program blocks, block number-Part Programming Examples-N-sequence No, G-preparatory Function, X,Y,Z – Co-ordinates- I,J,K – Co ordinates of centers, R – radius, F – feed, S – Spindle speed, M – miscellaneous function, EOB – End of the block-Programming Procedure For Manual NC Programming-Defining the machining operations-Necessary tools-Technological data-Geometric data-Generating NC program test.

MODULE III

Effect of Tool Nose Radius-Variou Turning Tools-Compensation, G41, G42-Cancel Compensation-Fundamentals of Machining Cycles (CNC Fixed/Canned Cycles)-Definition-Purpose-Format, Program Examples-Programme Entry And Execution-Programme Editing-Checking Syntax-Execution, Simulation-Tool Path.

MODULE IV

Machine Axes, 4axis, 5axis-Machine Zero, Work Zero, Reference Zero- Zero Offset-Length Compensation-G Codes & M Codes Explanations-Plane Selection-Point-Point Machining -Contour Milling-Linear Interpolation, Circular Interpolation, G01/G02/G03, IJKR- Programming Examples Using Machining Cycles (CNC Fixed/Canned Cycles)- Drilling, Tapping, Boring-Drilling Cycles G73, G81,G82, G83-Programming Examples-Canned Cycle Cancel G80- Z Return Code G99, G98-Dwell -Cutter Radius Compensation- Effect Of Cutter Radius On Contour-CRC Left And Right -Tool Length Compensation-Boring And Tapping Cycles-Format, R & L Hand Thread-Programming Examples-Milling Rectangular Pockets-Cycles, Allowances, Position Of Pocket, Selection Of Starting Corner, Feed Rate, Spindle Speed-Programming Examples-Milling Circular Pockets-Milling Direction, Pocket, Radius, Feed Rate, Spindle Speed-Finishing Allowance -Feed Rate, Spindle Speed, Milling Direction-Minor Diameter, Compensation- Mirror Image-Definition-Prerequisites-Programming Using M98, M99,M70, M71,M80 & M81.

REFERENCE

1. CNC Programming - Dr. S K Sinha - Galgotia Publications, New Delhi.
2. Computer Numerical Control Machine – Radhakrishnan P – New Central Book Agency
3. Computer Numerical Control of Machine Tools – Thyer GE – BH Newberg
4. C.N.C. Technology & Programming - Krar S - Hc Graw Hill